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BELAGAVI-590 018



A REPORT ON MINI-PROJECT WORK

IoT BASED FALL DETECTION AND ALERT SYSTEM

Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS & COMMUNICATION

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MINI PROJECT WORK (21ECMP67)

Course Learning Objectives:

This course will enable us to:

- Understand and analyze an engineering problem.
- Acquire technical knowledge and collect the information.
- Enhance communication, technical presentation and report preparation skills.
- Provide an opportunity to exercise the creative and innovative ideas in group.

ABSTRACT

This project proposes an IoT-based Fall Detection and Alert System that utilizes sensors and wireless communication to detect falls and alert caregivers or emergency services in real-time. The system consists of a wearable sensor device that detects changes in acceleration and orientation, a microcontroller that processes the data and triggers an alert, and a cloud-based platform that receives and dispatches alerts to designated contacts. The system aims to improve response times and reduce false alarms, enhancing the safety and well-being of individuals at risk of falls. Experimental results show high accuracy in fall detection and efficient alert transmission, demonstrating the system's potential to improve healthcare outcomes and quality of life.

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CHAPTER 1

INTRODUCTION

Falls are a major health concern, particularly for the elderly and individuals with mobility issues. The consequences of falls can be severe, ranging from minor injuries to fatal outcomes. Early detection and timely response are crucial in preventing long-term damage and improving outcomes. The Internet of Things (IoT) has revolutionized healthcare by enabling real-time monitoring and detection of falls.

This IoT-based Fall Detection and Alert System leverages advanced sensors, machine learning algorithms, and wireless connectivity to detect falls accurately and recognize patterns to prevent future falls. Our system aims to:

- Detect falls in real-time
- Recognize fall patterns and predict potential falls
- Alert caregivers and emergency services instantly
- Provide personalized insights for fall prevention strategies
- Enhance safety, independence, and quality of life

By harnessing the power of IoT technology, our system enables proactive fall management, improving healthcare outcomes and transforming lives.

CHAPTER 2

OBJECTIVES

The objectives of an IoT-based Fall Detection and Alert system can be categorized into two main areas:

1. Reduced Fall-related Injuries:

- **Importance of Early Intervention:** Early assistance can prevent:
 - **Secondary complications:** Lying on the floor for extended periods can lead to pressure sores, pneumonia, or dehydration.
 - **Further injury:** An incapacitated individual might fall again or injure themselves further trying to get up.
- **Benefits of Reduced Injuries:**
 - Improved health outcomes for at-risk individuals.
 - Reduced healthcare costs associated with fall-related injuries.

2. Improved Independence for At-Risk Individuals:

- **Psychological Impact:** Knowing there's a safety net in place can:
 - Reduce anxiety and fear of falling.
 - Increase confidence in performing daily activities.
 - Promote a sense of control and autonomy.
- **Maintaining Independence:**
 - The fall detection system allows individuals to:
 - Remain in their homes for longer.
 - Engage in daily activities without constant supervision.
 - Live a more fulfilling and independent life.

3. Peace of Mind for Caregivers and Families:

- **Reduced Stress and Anxiety:** Knowing their loved ones are protected provides caregivers with:
 - A sense of security and control.
 - Peace of mind in knowing help will arrive quickly in case of a fall.

- Reduced worry about potential accidents.

CHAPTER 3

3.1 BLOCK DIAGRAM

METHODOLOGY

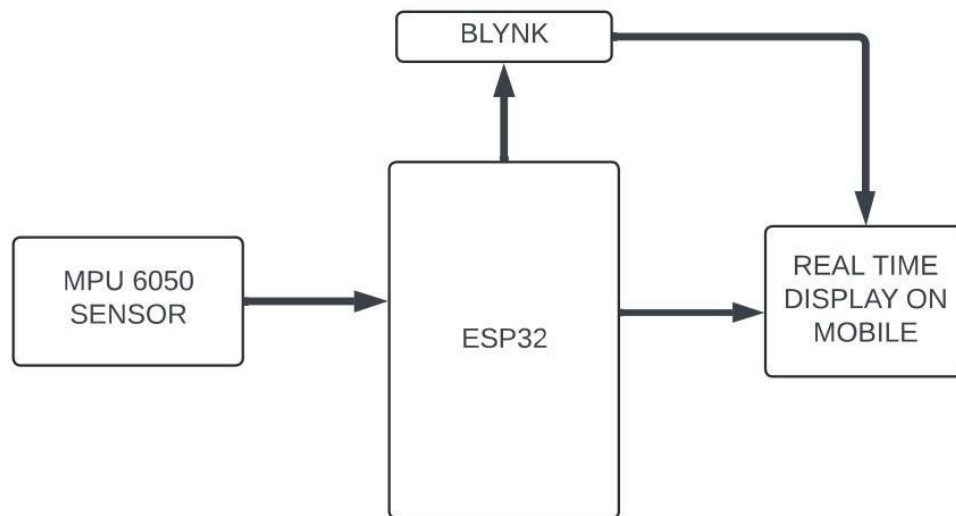


Figure 3.1.1 Block Diagram

3.1.1 BLOCK DIAGRAM DESCRIPTION:

The block depicts overall working of an IoT-based fall detection system. Here's a breakdown of the system's components:

- **MPU 6050 Sensor:** This sensor measures motion in three dimensions: X, Y, and Z. It combines an accelerometer and a gyroscope. The accelerometer detects acceleration due to gravity and motion, while the gyroscope detects rotational motion or angular velocity.
- **ESP32 microcontroller:** This microcontroller reads data from the MPU 6050 sensor and transmits it for processing and analysis.
- **BLYNK:** This is an IoT platform used for data visualization, device management, and app creation. It likely receives the data from the ESP32 microcontroller and sends fall alerts to a smartphone app.
- **Real-time Display on Mobile:** This refers to the fall alert notification that would be received on a smartphone app.

3.2 FLOW CHART

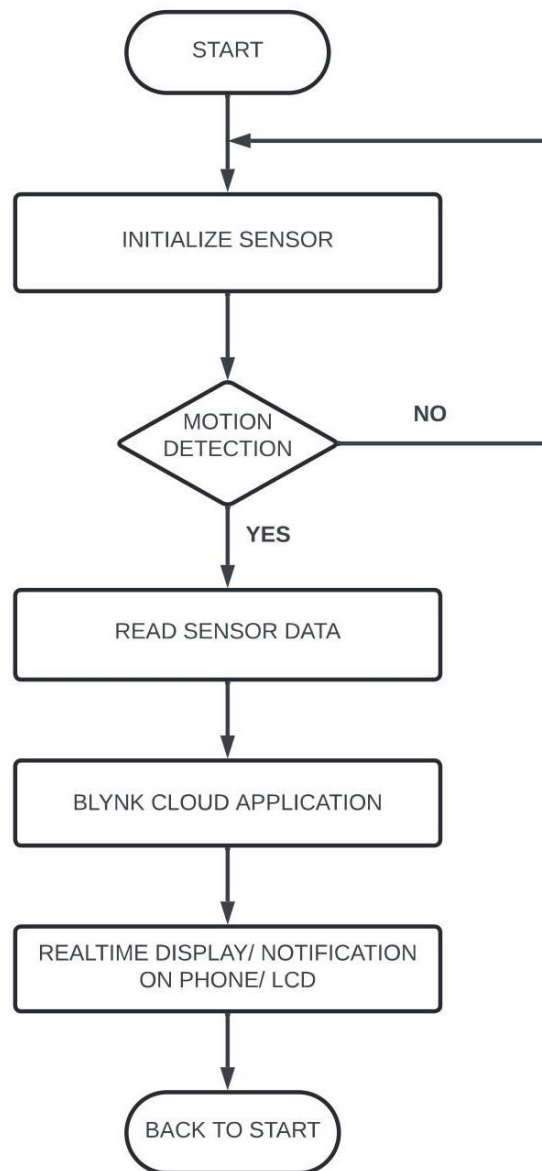


Figure 3.2.1 Flow chart

CHAPTER 4

HARDWARE DESCRIPTION

4.1 ESP32 WROOM Model



Figure 4.1.1 ESP32

- ESP32-WROOM-32 is a powerful, generic Wi-Fi + Bluetooth® + Bluetooth LE MCU module that targets a wide variety of applications, ranging from low-power sensor networks to the most demanding tasks, such as voice encoding, music streaming and MP3 decoding.
- The integration of Bluetooth, Bluetooth LE and Wi-Fi ensures that a wide range of applications can be targeted, and that the module is all-around: using Wi-Fi allows a large physical range and direct connection to the Internet through a Wi-Fi router, while using Bluetooth allows the user to conveniently connect to the phone or broadcast low energy beacons for its detection.

Core Functionality:

- **Microcontroller Unit (MCU):** It incorporates a dual-core Xtensa® 32-bit LX6 microprocessor. These cores can be individually controlled or powered off for optimized performance depending on the application's needs. The clock frequency is adjustable from 80 MHz to 240 MHz, allowing for flexibility in processing power.
- **Wireless Connectivity:**

Wi-Fi: Supports Wi-Fi protocols like 802.11 b/g/n, enabling connection to local Wi-Fi networks for internet access or data transfer.

- **Bluetooth:** Offers classic Bluetooth capabilities for legacy device connections, supporting profiles like L2CAP, SDP, GAP, SMP, and more.
- **Bluetooth Low Energy (BLE):** Supports BLE for connecting to low-power devices like smartphones and wearables. This technology is ideal for battery-powered applications as it consumes less energy compared to classic Bluetooth.

Additional Features:

- **Integrated Onboard Memory:** The ESP32 WROOM module typically includes built-in flash memory, allowing for program and data storage.
- **Rich Peripheral Set:** It offers a variety of peripherals like GPIO pins, ADC (Analog-to-Digital Converter), SPI (Serial Peripheral Interface), I2C (Inter-Integrated Circuit), and more. These features enable interaction with various sensors, displays, and other external components.
- **Low Power Consumption:** Designed with power efficiency in mind, the ESP32 WROOM module can operate in different power modes, allowing developers to optimize battery life for specific applications.
- **Compact Size:** The module is relatively small and lightweight, making it suitable for integration into space-constrained projects.

Applications:

The ESP32 WROOM module's versatility makes it ideal for various IoT (Internet of Things) applications, including:

- **Wearable electronics:** Smartwatches, fitness trackers, health monitoring devices.
- **Smart home devices:** Thermostats, light switches, security systems.
- **Industrial automation:** Sensor data collection, control systems, monitoring equipment.
- **Robotics:** Wireless control, data transmission.
- **Wireless communication projects:** Data logging, remote control systems.

4.2 MPU 6050

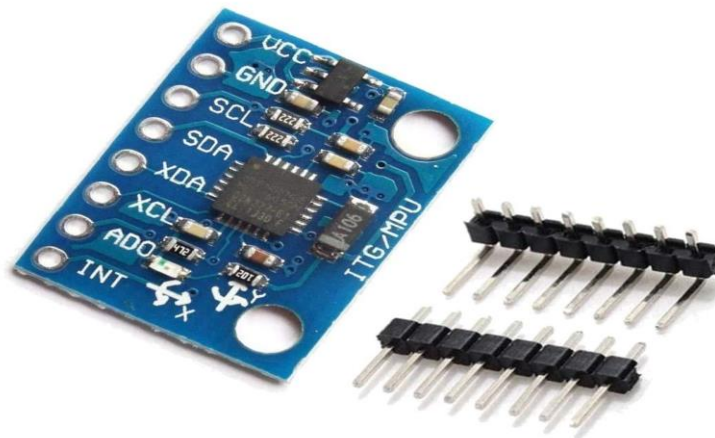


Figure 4.2.1 MPU6050 SENSOR

The MPU6050 is a popular **six-axis Motion Tracking Device (MTD)** widely used in various applications that require motion or orientation detection. The key features of the sensor are:

Sensor Combination:

- The MPU6050 integrates two critical sensors on a single chip:
 - **3-axis Accelerometer:** This sensor detects acceleration due to gravity and motion along the X, Y, and Z axes. It helps determine the device's tilt or inclination.
 - **3-axis Gyroscope:** This sensor measures the device's rotational velocity or angular speed around the X, Y, and Z axes. This helps track changes in orientation.

Functionality:

- By combining the data from both the accelerometer and gyroscope, the MPU6050 provides a more comprehensive picture of the device's motion and orientation in 3D space.
- It utilizes **Micro-Electro-Mechanical Systems (MEMS)** technology, where microscopic mechanical elements and electronic circuits are integrated onto a single chip.
- The sensor offers various configurable settings for:
 - **Full-scale acceleration range:** This can be set to different values (e.g., +/- 2g, +/- 4g, +/- 8g, +/- 16g) depending on the application's needs. "g" refers to the force of gravity.

- **Gyroscope sensitivity:** This determines the range of angular velocities the sensor can measure.

Additional Features:

- **Digital Motion Processor (DMP):** Some versions of the MPU6050 include an onboard DMP that can perform complex motion calculations internally, reducing the processing load on the main microcontroller.
- **Temperature Sensor:** Incorporates a built-in temperature sensor for additional environmental data.
- **I2C Communication Interface:** The MPU6050 uses a standard I2C (Inter-Integrated Circuit) communication protocol to interact with microcontrollers, simplifying data exchange.

Applications:

The MPU6050's ability to sense motion and orientation makes it suitable for various applications, including:

- **Wearable electronics:** Smartwatches, fitness trackers, virtual reality headsets.
- **Robotics:** Balance control, motion detection, object manipulation.
- **Gaming controllers:** Motion-sensing features in gamepads.
- **Drone and UAV control:** Flight stabilization, orientation tracking.
- **Internet of Things (IoT) devices:** Motion-activated devices, gesture recognition.

Overall, the MPU6050 is a compact, affordable, and versatile sensor that provides valuable motion and orientation data for various projects involving movement or position tracking.

4.3 Circuit Diagram

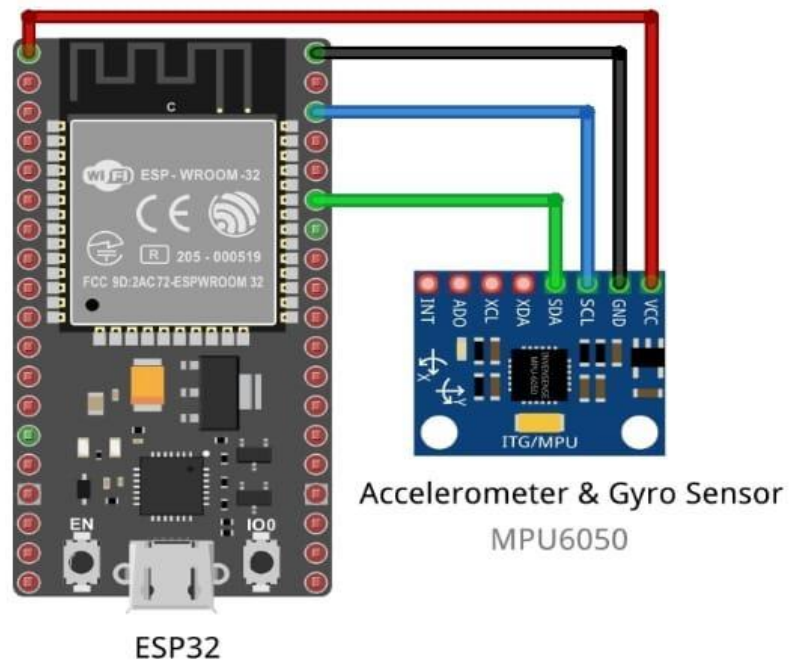


Figure 4.3.1 Circuit Diagram

CHAPTER 5

SOFTWARE REQUIREMENT

5.1 Arduino IDE



Figure 5.1.1 Arduino IDE

Arduino uno for programming Arduino microcontroller. Arduino uno can be implemented within Windows, Mac and Linux operating systems. Most of its components are written in c language for easy editing and compiling. Users can modify internal layouts and schematics when required.

Integrated Development Environment “IDE” For Arduino

Arduino IDE is an open-source software that is mainly used for writing and compiling the code into the Arduino Module.

It is an official Arduino software, making code compilation too easy that even a common person with no prior technical knowledge can get their feet wet with the learning process. It is easily available for operating systems like MAC, Windows, Linux and runs on the Java Platform that comes with inbuilt functions and commands that play a vital role for debugging, editing and compiling the code in the environment.

Tools

Auto Format This formats your code nicely: i.e., indents it so that opening and closing curly braces line up, and that the statements inside curly braces are indented more. **Archive Sketch** Archives a copy of the current sketch in .zip format. The archive is placed in the same directory as the sketch. **Fix Encoding & Reload** Fixes possible discrepancies between the editor char map encoding and other operating systems char maps. **Serial Monitor** Opens the serial monitor window and initiates the exchange of data with any

connected board on the currently selected Port. This usually resets the board, if the board supports Reset over serial port opening.

Board Select the board that you're using. See below for descriptions of the various boards. **Port** This menu contains all the serial devices (real or virtual) on your machine. It should automatically refresh every time you open the top-level tools menu. **Programmer** For selecting a hardware programmer when programming a board or chip and not using the onboard USB serial connection. Normally you won't need this, but if you're burning a bootloader to a new microcontroller, you will use this. **Burn Bootloader** The items in this menu allow you to burn a bootloader onto the microcontroller on an Arduino board. This is not required for normal use of an Arduino board but is useful if you purchase a new ATmega microcontroller (which normally come without a bootloader). Ensure that you've selected the correct board from the Boards menu before burning the bootloader on the target board. This command also set the right fuses.

Writing Sketches

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension. `.ino`. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor. Arduino IDE is used to control all the input and output devices that are connected to the Arduino uno with programming.

For code, the following libraries are installed:

WiFi.h: Handles Wi-Fi connectivity on the ESP32.

WiFiClient.h: Used for creating a Wi-Fi client connection.

BlynkSimpleEsp32.h: Enables communication with the Blynk IoT platform on the ESP32.

Adafruit_MPU6050.h: Provides support for the MPU6050 sensor, allowing reading of the data from the sensor.

Adafruit_Sensor.h: Includes sensor event data types and helper functions used with the Adafruit sensor libraries.

Wire.h: For I2C communication, used for connecting and communicating with the MPU6050 sensor.

5.2 BLYNK CLOUD



Figure 5.2.1 Blynk Icon

Blynk Cloud is a cloud-based platform designed to simplify the development and management of Internet of Things (IoT) projects. It offers a suite of tools for connecting hardware devices to mobile applications and the internet, allowing for remote control, data visualization, and project management. Here's a breakdown of its key features:

- **Device Management:**

- Blynk Cloud facilitates the connection and management of various IoT hardware devices, including popular boards like Arduino, Raspberry Pi, and ESP32.
- It provides a user-friendly interface for configuring devices and associating them with specific projects.

- **Mobile App Development:**

- Blynk offers a visual programming environment for building user interfaces on smartphones and tablets. This drag-and-drop interface allows users to create custom dashboards featuring various widgets like buttons, sliders, graphs, and displays.
- These widgets can be linked to specific hardware functions, enabling users to control devices remotely from their mobile apps.

- **Data Visualization and Storage:**

- Blynk Cloud enables the visualization of sensor data collected from connected devices. Users can create graphs and charts to monitor sensor readings in real-time or analyse historical data.
- The platform might offer limited data storage capabilities, allowing users to retain sensor data for later analysis.

- **Real-time Communication:**

- Blynk Cloud facilitates real-time communication between devices and mobile apps. Users can trigger actions on devices based on app interactions or send commands based on sensor readings.

- **Security:**

- Blynk Cloud implements security measures to protect user data and device communication. This might include features like user authentication and secure communication protocols.

Benefits of using Blynk Cloud:

- **Simplified Development:** Blynk Cloud's visual programming approach simplifies the creation of mobile apps for IoT projects, even for users with limited coding experience.
- **Rapid Prototyping:** The platform allows for quick prototyping of IoT projects without requiring extensive coding efforts.
- **Remote Control and Monitoring:** Users can remotely control and monitor their IoT devices from anywhere with an internet connection.
- **Scalability:** Blynk Cloud can support a range of projects, from simple hobbyist applications to more complex industrial deployments (depending on the chosen plan).

Limitations:

- **Limited Features:** The free tier of Blynk Cloud might have limitations on data storage, connected devices, or project complexity.
- **Cloud Dependence:** Projects rely on Blynk's cloud infrastructure for communication, which could introduce a single point of failure if the service becomes unavailable.
- **Security Considerations:** While Blynk implements security measures, users should be aware of potential security risks associated with cloud-based platforms.

Overall, Blynk Cloud offers a valuable tool for developers and hobbyists to create and manage IoT projects. Its user-friendly interface and suite of features make it a popular choice for rapid prototyping and remote-control applications.

CHAPTER 6

RESULT AND DISCUSSION

The results of this project IoT based fall detection and alert system are discussed in the following cases in detailed manner as explained in the methodology.

After uploading the code, open the monitor. The Monitor will display the initialization message.

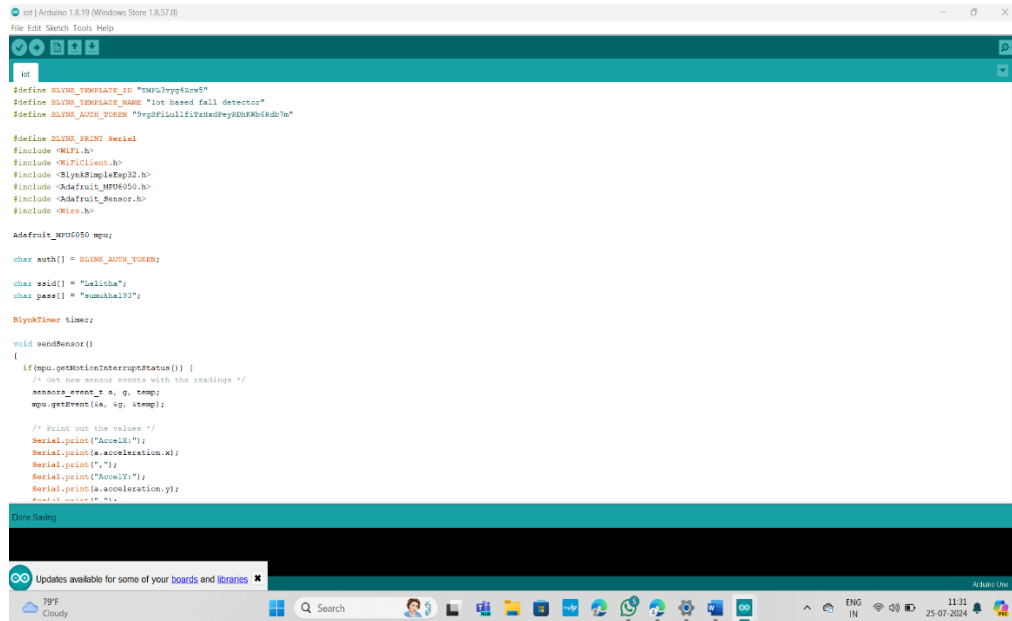


Figure 6.1.1 Arduino IDE software with Embedded C code and Initialization

If the position of the person changes then the family members/care takers get the notification in Blynk app in mobile and even they will get email notification. Below figure 6.2 Shows the circuit connections and mobile app notification.

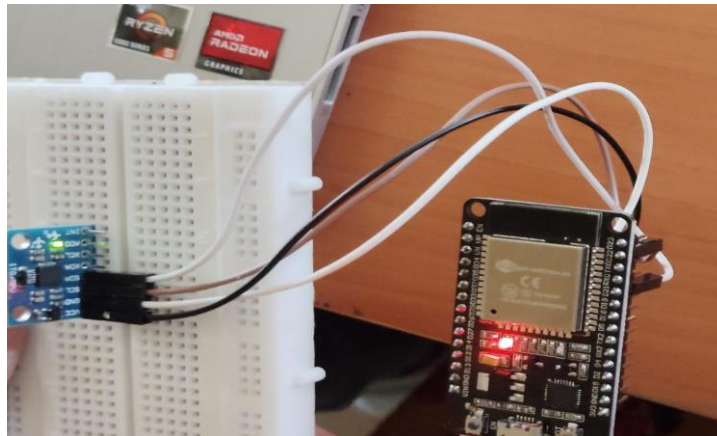


Figure 6.1.2 Interfacing of ESP32 with MPU6050

The acceleration of x, y and z, gyro x, y and z values will be displayed on serial monitor.

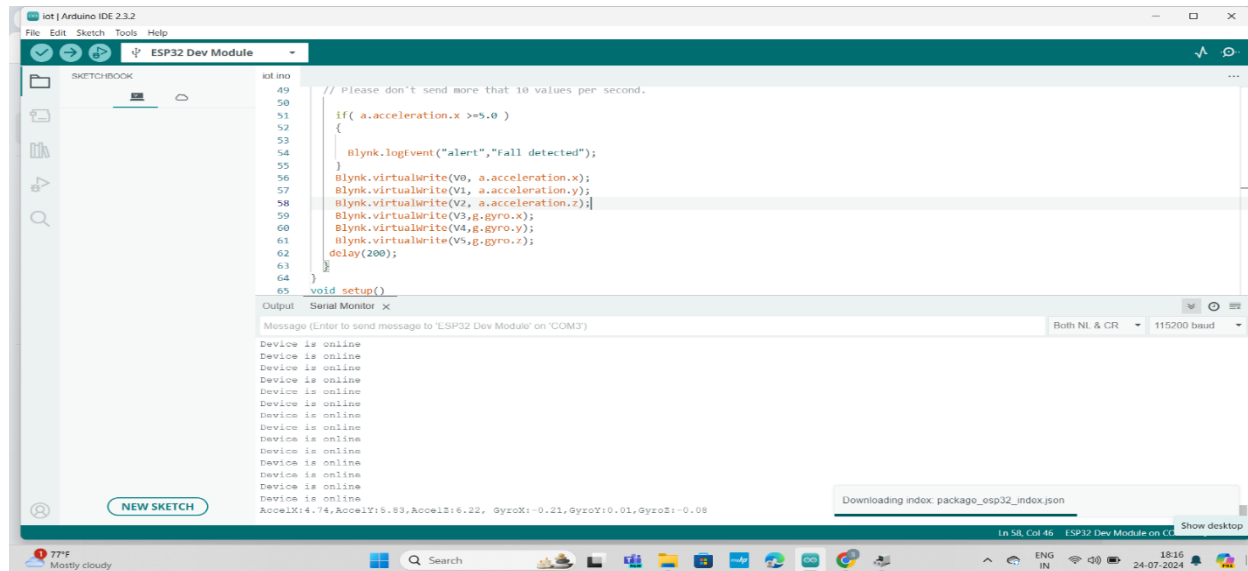


Figure 6.1.3 Serial Monitor Display

Blynk IoT Templates of Accelerometer and Gyroscope.

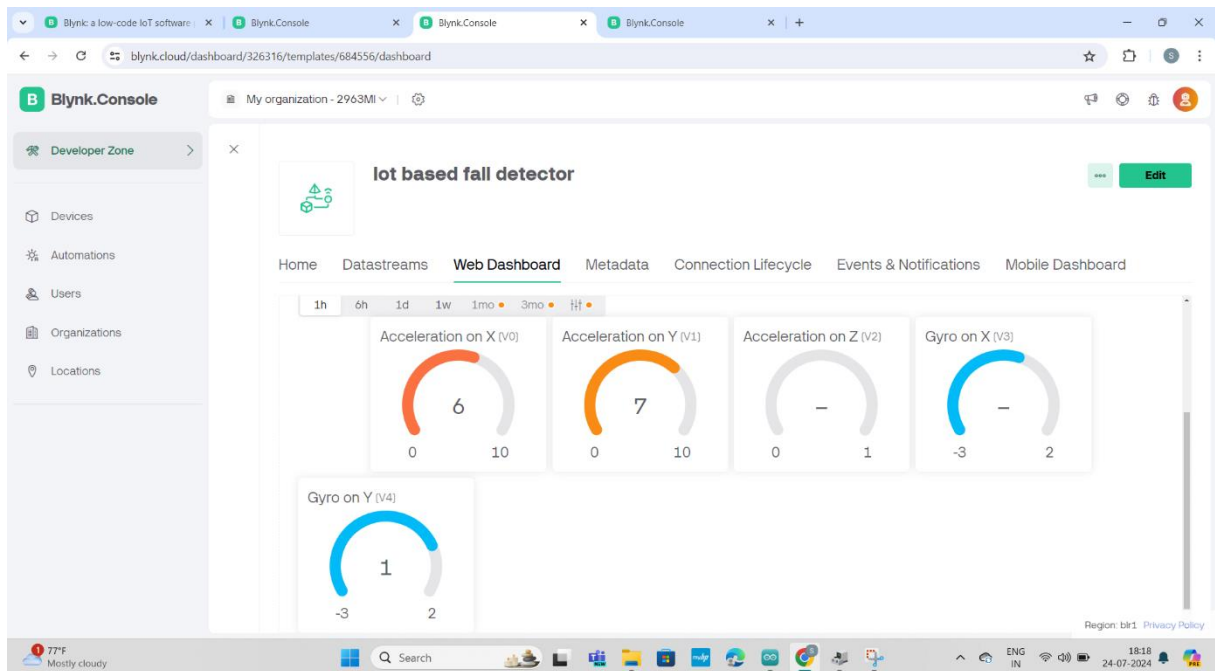


Figure 6.1.4 Blynk dashboard

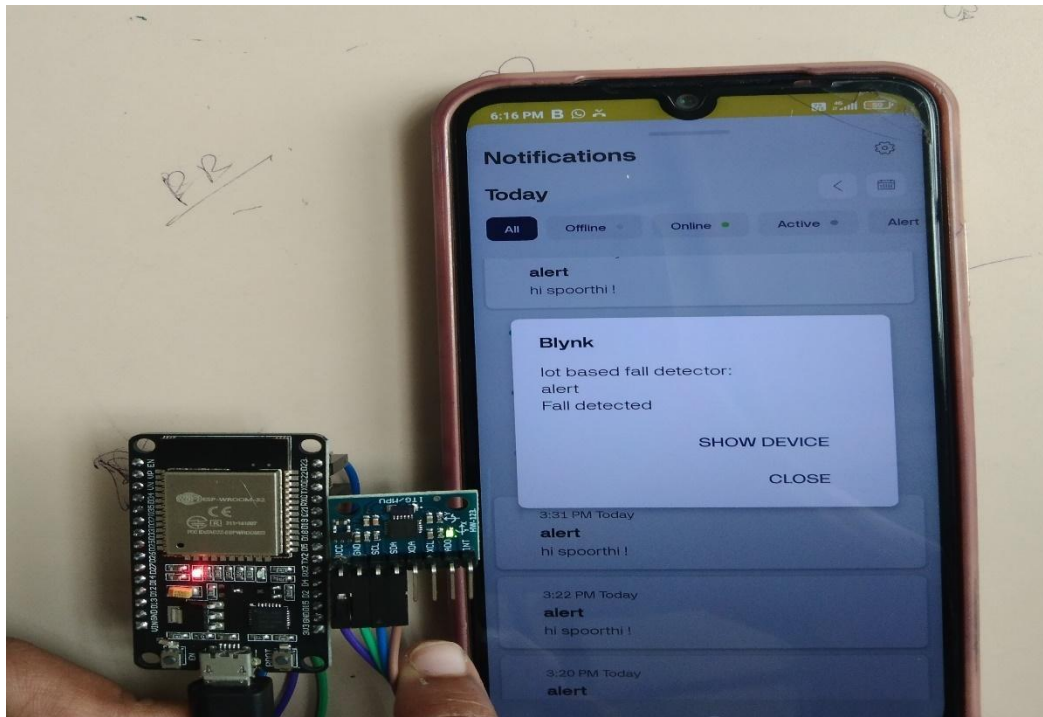


Figure 6.1.5 Blynk app notification on mobile

CHAPTER 7

APPLICATIONS

IoT-based fall detection and recognition systems offer a wide range of applications, particularly in scenarios where falls pose a significant risk. Here are some key areas where these systems can be beneficial:

1. Elderly Care:

- **Independent Living Support:** Allows seniors to live independently for longer by providing immediate assistance in case of a fall.
- **Remote Monitoring:** Enables caregivers to remotely monitor the well-being of elderly individuals living alone, offering peace of mind and facilitating timely intervention in case of a fall.
- **Improved Quality of Life:** Provides a sense of security and reduces anxiety for both seniors and their families.

2. Healthcare Facilities:

- **Patient Monitoring:** Helps healthcare personnel monitor fall-prone patients in hospitals, nursing homes, and rehabilitation centres, ensuring prompt assistance in case of a fall incident.
- **Reduced Staff Burden:** Frees up staff time from constant monitoring by triggering alerts only when needed.
- **Improved Patient Outcomes:** Faster response to falls can potentially minimize injuries and improve patient recovery.

3. Assisted Living Facilities:

- **Enhanced Safety for Residents:** Provides an extra layer of protection for residents with a higher risk of falls, offering peace of mind to both residents and their families.
- **Improved Staff Efficiency:** Allows staff to focus on providing care and support while the system monitors residents for falls.

4. Industrial Settings:

- **Worker Safety:** Protects workers in high-risk environments (e.g., construction sites) by triggering alerts for falls that might otherwise go unnoticed.
- **Reduced Work Injuries:** Faster response to falls can potentially decrease the severity of injuries and workers' compensation claims.

IoT based Fall Detection and Alert System

- **Data-Driven Insights:** Collected data on falls can be analysed to identify areas for improvement in workplace safety protocols.

ADVANTAGES

Main advantages of our project are:

- **Improved Safety and Reduced Risks:** This is the most significant advantage. By promptly detecting falls and triggering alerts, the system can ensure faster response times for medical attention. This can potentially minimize injuries, prevent complications, and even save lives.
- **Enhanced Independence for Elderly and At-Risk Individuals:** The system empowers individuals with a higher fall risk to live more independently. Knowing they have a safety net in place can reduce anxiety and allow them to maintain a sense of autonomy.
- **Peace of Mind for Caregivers and Families:** The system provides peace of mind for caregivers and families living remotely from their loved ones. They can be assured that help will be on the way if a fall occurs.
- **Reduced Healthcare Costs:** Faster intervention after a fall can potentially lead to shorter hospital stays, fewer complications, and lower overall healthcare costs.
- **Improved Staff Efficiency in Care Settings:** By automating fall detection, the system frees up staff time in healthcare facilities and assisted living centres. Staff can focus on providing care and attention to residents who need it most.
- **Non-Intrusive and User-Friendly:** Many IoT fall detection systems utilize wearable devices or sensors that are comfortable and non-intrusive. This makes the technology more readily accepted by users.
- **Scalability and Adaptability:** These systems can be tailored to different needs and environments. Some systems use wearables, while others rely on in-home sensors, offering flexibility depending on the situation.

Overall, IoT-based Fall Detection and Alert systems offer a valuable tool for promoting safety, independence, and peace of mind for individuals at risk of falls and their loved ones.

LIMITATIONS

The limitations of IoT-based Fall Detection and Alert systems:

Accuracy and False Alarms:

- **Distinguishing Falls from Daily Activities:** The system might struggle to differentiate between actual falls and daily activities that involve sudden movements (e.g., bending down, tripping). This can lead to false alarms, causing unnecessary worry and wasting resources.
- **Sensor Limitations:** Sensor data might be affected by body position, clothing, or environmental factors, potentially leading to missed falls or inaccurate interpretations.

User Acceptance and Compliance:

- **Wearable Comfort and User Preference:** Not everyone enjoys wearing a device 24/7. Some users might find wearables uncomfortable or inconvenient, leading to non-compliance.
- **In-Home Sensor Placement Limitations:** Sensor placement in a home environment might be limited by furniture or room layout, potentially creating blind spots where falls cannot be detected.

Technical Challenges:

- **Battery Life:** Wearable devices rely on batteries, and frequent recharging can be a hassle for users. Low battery can lead to system downtime and missed fall events.
- **Data Security and Privacy:** The system collects user data, raising concerns about data security and privacy breaches. Robust security measures are essential to protect sensitive data.
- **Cost Considerations:** The cost of the system itself (devices, subscriptions) might be a barrier for some users. Additionally, potential false alarms could lead to extra healthcare costs.
- **System Complexity:** Setting up and maintaining the system might require some technical knowledge, posing challenges for users with limited technical expertise.

CONCLUSION

An IoT-based Fall Detection and Alert system offers a promising solution for promoting safety and well-being for individuals at risk of falls. The system boasts significant advantages like improved fall response times, increased independence for at-risk individuals, and peace of mind for caregivers.

However, limitations like potential for false alarms, user compliance issues, and technical challenges require further development.

Despite these limitations, the potential benefits of this technology are undeniable. With continuous advancements in sensor technology, data processing algorithms, and user-friendly design, IoT-based fall detection systems have the potential to become a widespread tool for safeguarding individuals and enhancing overall well-being.

FUTURE SCOPE

The future of IoT-based fall detection and recognition systems holds exciting possibilities for enhancing safety and promoting well-being. Here are some key areas of exploration:

1. Advanced Fall Detection Algorithms:

- Machine learning algorithms can be further refined to differentiate falls from daily activities more accurately, minimizing false alarms.
- Integration with additional sensors (e.g., pressure sensors, microphones) could provide a more comprehensive picture of a fall event, leading to more precise detection.

2. Integration with Smart Homes and Wearables:

- Seamless integration with smart home features like smart lights or automatic door unlocking could provide immediate assistance after a fall.
- Advanced wearables that monitor vital signs (e.g., heart rate, blood oxygen) could trigger emergency services based on physiological changes during a fall.

3. Improved User Experience:

- Development of smaller, lighter, and more comfortable wearables will enhance user acceptance and compliance.
- The system could offer personalized settings and user interfaces, catering to individual needs and preferences.
- Integration with voice assistants for hands-free interaction might improve accessibility.

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APPENDIX A

DATASHEET

DATA SHEET OF ESP32 WROOM 32

Category	Items	Specification
Microcontroller	Cores	Xtensa LX6 (dual-core)
	Clock Frequency	80 MHz to 240 MHz
Memory	Flash	32 Mbits SPI flash
Wireless	Wi-Fi	802.11 b/g/n, up to 150 Mbps
	Bluetooth	v4.2 BR/EDR, BLE
Interfaces	UART	Yes
	SPI	Yes
	I2C	Yes
	SD Card	Yes
	Other	Capacitive touch sensor, ADC, DAC, GPIO, and more
Power Supply	Voltage	3.3V
Power Consumption	Sleep Current	< 5 μ A
Dimensions	(L x W x H)	18 mm x 25.5 mm x 3.1 mm

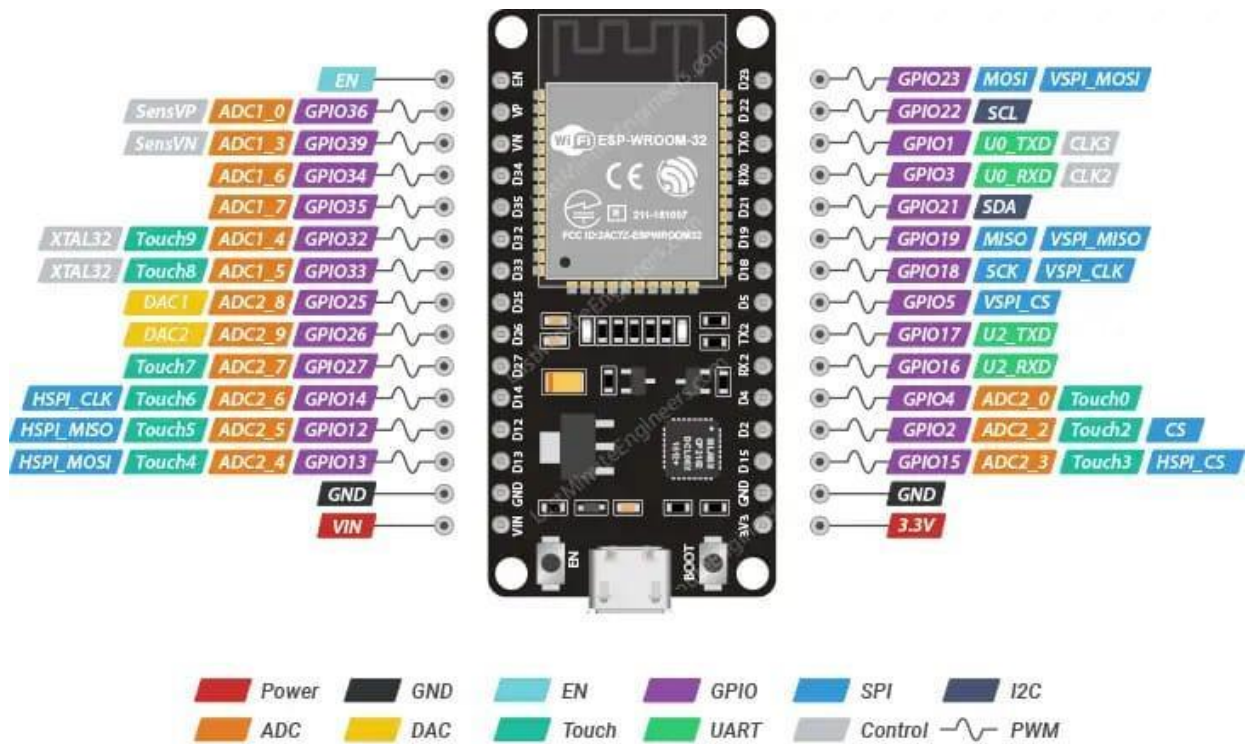


Figure: ESP32 Pin Configuration

DATA SHEET OF MPU6050

Part/Item	MPU6050
VDD	2.37- 3.46V
VLOGIC	1.71V to VDD
Serial Interfaces Supported	I ² C
Pin 8	VLOGIC
Pin 9	AD0
Pin 23	SCL
Pin 24	SDA

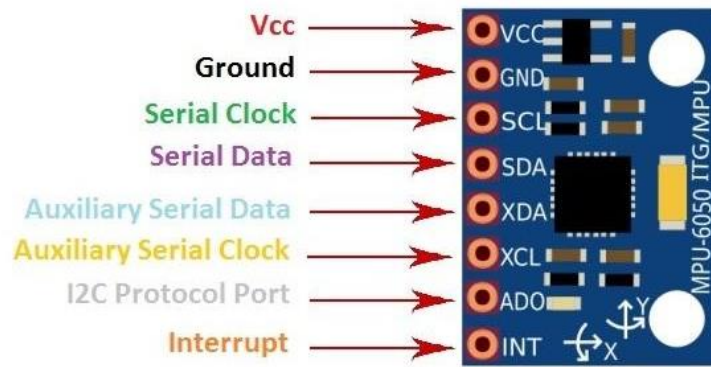


Figure: Pin Configuration of MPU6050 sensor

B: SOURCE CODE

```
#define BLYNK_TEMPLATE_ID "TMPL3vyg4Zcw5"

#define BLYNK_TEMPLATE_NAME "Iot based fall detector"

#define BLYNK_AUTH_TOKEN "9vgSFilullfiTzHxdFeyRDhKWb6Rdb7m"

#define BLYNK_PRINT Serial

#include <WiFi.h>

#include <WiFiClient.h>

#include <BlynkSimpleEsp32.h>

#include <Adafruit_MPU6050.h>

#include <Adafruit_Sensor.h>

#include <Wire.h>

Adafruit_MPU6050 mpu;

char auth[] = BLYNK_AUTH_TOKEN;

char ssid[] = "sindhu";

char pass[] = "sindhu193";

BlynkTimer timer;

void sendSensor()

{

if(mpu.getMotionInterruptStatus()) {

/* Get new sensor events with the readings */

sensors_event_t a, g, temp;

mpu.getEvent(&a, &g, &temp);

/* Print out the values */

Serial.print("AccelX:");
```

```
Serial.print(a.acceleration.x);

Serial.print(", ");

Serial.print("AccelY:");

Serial.print(a.acceleration.y);

Serial.print(", ");

Serial.print("AccelZ:");

Serial.print(a.acceleration.z);

Serial.print(", ");

Serial.print("GyroX:");

Serial.print(g.gyro.x);

Serial.print(",");

Serial.print("GyroY:");

Serial.print(g.gyro.y);

Serial.print(",");

Serial.print("GyroZ:");

Serial.print(g.gyro.z);

Serial.println("");

// You can send any value at any time.

// Please don't send more that 10 values per second.

    if( a.acceleration.x >=5.0 )

    {

        Blynk.logEvent("alert", "Fall detected");

    }

Blynk.virtualWrite(V0, a.acceleration.x);

Blynk.virtualWrite(V1, a.acceleration.y);
```

```
Blynk.virtualWrite(V2, a.acceleration.z);

Blynk.virtualWrite(V3,g.gyro.x);

Blynk.virtualWrite(V4,g.gyro.y);

Blynk.virtualWrite(V5,g.gyro.z);

delay(200);

}

}

void setup()

{

  Serial.begin(115200);

  Blynk.begin(auth, ssid, pass);

  Blynk.logEvent("alert","Fall detected !");

  while (!Serial)

    delay(10); // will pause Zero, Leonardo, etc until serial console opens

  Serial.println("Adafruit MPU6050 test!");

  // Try to initialize!

  if (!mpu.begin()) {

    Serial.println("Failed to find MPU6050 chip");

    while (1) {

      delay(10);

    }

  }

  Serial.println("MPU6050 Found!");

  //setupt motion detection

  mpu.setHighPassFilter(MPU6050_HIGHPASS_0_63_HZ);
```

```
mpu.setMotionDetectionThreshold(1);

mpu.setMotionDetectionDuration(20);

mpu.setInterruptPinLatch(true); // Keep it latched. Will turn off when reinitialized.

mpu.setInterruptPinPolarity(true);

mpu.setMotionInterrupt(true);

Serial.println("");

delay(100);

timer.setInterval(1000L, sendSensor);

}

void loop()

{

  if(Blynk.connected())

  {

    Serial.println("Device is online");

  }

  else

  {

    Serial.println("Device is offline")

  }

  Blynk.run();

  timer.run();

}
```

Course Outcomes

On completion of this course, we are able to

- Solve the identified problems.
- Analyze the available resources and their utilization.
- Present the work carried out and prepare the report.
- Work in a team to find the solutions for societal and technical problems.

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